

```

private void InitializeComponent()
{
    t1= new TextBox();
    t1.Location = new System.Drawing.Point(100, 100 );
    t1.Size = new System.Drawing.Size(200, 0);

    t2= new TextBox();
    t2.Location = new System.Drawing.Point(100, 150 );
    t2.Size = new System.Drawing.Size(200, 0);
    t3= new TextBox();
    t3.Location = new System.Drawing.Point(100, 250 );
    t3.Size = new System.Drawing.Size(200, 0);

    BtnAdd = new Button();
    BtnAdd.Location = new System.Drawing.Point(100, 200);
    BtnAdd.Size = new System.Drawing.Size(200, 30);
    BtnAdd.Text = "Add";
    BtnAdd.Click += new System.EventHandler(this.BtnAdd_Click);

    b1 = new Button();
    b1.Location = new System.Drawing.Point(100, 300);
    b1.Size = new System.Drawing.Size(200, 30);
    b1.Text = "B1";
    b1.Click += new System.EventHandler(this.BtnHello_Click);
    b2 = new Button();
    b2.Location = new System.Drawing.Point(100, 350);
    b2.Size = new System.Drawing.Size(200, 30);
    b2.Text = "B2";
    b2.Click += new System.EventHandler(this.BtnHi_Click);
    b2.Click += new System.EventHandler(this.BtnHello_Click);
    b2.Click -= new System.EventHandler(this.BtnHi_Click);

    this.Controls.Add(t1);
    this.Controls.Add(t2);
    this.Controls.Add(t3);
    this.Controls.Add(BtnAdd);
    this.Controls.Add(b1);
    this.Controls.Add(b2);
    this.components = new System.ComponentModel.Container();
    this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
    this.ClientSize = new System.Drawing.Size(800, 450);
    this.Text = "Form1";
    this.WindowState = System.Windows.Forms.FormWindowState.Maximized;
}

```

5 references | 5 references | 5 references

private TextBox t1,t2,t3;

6 references | 6 references | 8 references

Button BtnAdd,b1,b2;

#endregion

```
1 namespace WinFormApp;
  3 references
2 public partial class Form1 : Form
3 {
    1 reference
4     public Form1()
5     {
6         InitializeComponent();
7     }
    1 reference
8     public void BtnAdd_Click(object sender, EventArgs e)
9     {
10
11         t3.Text = (Convert.ToInt16(t1.Text) +int.Parse(t2.Text)) .ToString();
12     }
    2 references
13     public void BtnHello_Click(object sender, EventArgs e)
14     {
15         MessageBox.Show("Hello World");
16     }
    2 references
17     public void BtnHi_Click(object sender, EventArgs e)
18     {
19         MessageBox.Show("Hi World");
20     }
21 }
22
```

```
private void InitializeComponent()
```

```
{  
    this.components = new System.ComponentModel.Container();  
    this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;  
    this.ClientSize = new System.Drawing.Size(800, 450);  
    this.WindowState = System.Windows.Forms.FormWindowState.Maximized;  
    this.Text = "Form1";  
  
    t1=new TextBox();  
    t1.Location = new System.Drawing.Point(100,100);  
    t1.Size = new System.Drawing.Size(200,0);  
    // t1.Name= "t1";  
    t1.KeyPress += new System.Windows.Forms.KeyPressEventHandler(this.t1_keypress);  
    t1.TextChanged += new System.EventHandler(this.t1_textchanged);  
    t1.TabIndex=2;  
    this.Controls.Add(t1);  
  
    t2=new TextBox();  
    t2.Location = new System.Drawing.Point(100,150);  
    t2.Size = new System.Drawing.Size(200,0);  
    // t2.Name= "t2";  
    t2.Enabled= false ;  
    this.Controls.Add(t2);  
  
    t3=new TextBox();  
    t3.Location = new System.Drawing.Point(100,200);  
    t3.Size = new System.Drawing.Size(200,0);  
    t3.Name= "t3";  
    t3.GotFocus += new System.EventHandler(this.t3_onfocus);  
    t3.Leave += new System.EventHandler(this.t3_onleave);  
    t3.TabIndex = 0;  
    t3.KeyDown += new System.Windows.Forms.KeyEventHandler (this.t3_keydown);  
    this.Controls.Add(t3);  
  
    t4=new TextBox();  
    t4.Location = new System.Drawing.Point(100,250);  
    t4.Size = new System.Drawing.Size(200,0);  
    t4.Name= "t4";  
    t4.MouseEnter += new System.EventHandler(this.t4_mouseenter);  
    t4.MouseLeave += new System.EventHandler(this.t4_mouseleave);  
    t4.KeyUp += new System.Windows.Forms.KeyEventHandler (this.t4_keyup);  
    this.Controls.Add(t4);  
  
    p1=new PictureBox();  
    p1.Location = new System.Drawing.Point(100,350);  
    p1.Size = new System.Drawing.Size(400,400);  
    p1.Name= "p1";  
    p1.BackColor = System.Drawing.Color.White;  
    p1.MouseUp+=new System.Windows.Forms.MouseEventHandler(this.p1_mouseup);  
    p1.MouseDown+=new System.Windows.Forms.MouseEventHandler(this.p1_mousedown);  
    this.Controls.Add(p1);  
}
```

```
81     b1=new Button();
82     b1.Location = new System.Drawing.Point(100,300);
83     b1.Size = new System.Drawing.Size(200,40);
84     b1.Name= "b1";
85     b1.Text= "Button";
86     b1.MouseDown += new System.Windows.Forms.MouseEventHandler(this.b1_mouseclick);
87     b1.FlatStyle = System.Windows.Forms.FlatStyle.System;
88     this.Controls.Add(b1);
89 }
90 8 references
91 private TextBox t1;
92 7 references
93 private TextBox t2;
94 12 references
95 private TextBox t3;
96 11 references
97 private TextBox t4;
98 12 references
99 private PictureBox p1;
100 8 references
101 private Button b1;

#endregion
}
```

```

1 namespace WinFormApp;
  2 references
2 public partial class Form2 : Form
3 {
  2 references
4     private Point startPoint;
  3 references
5     private bool isDrawing = false;
  3 references
6     private Bitmap canvas;
  0 references
7     public Form2()
8     {
9         InitializeComponent();
10        canvas = new Bitmap(p1.Width, p1.Height);
11    }
  1 reference
12    public void t1_keypress(object sender, KeyPressEventArgs e)
13    {
14        t2.Text = e.KeyChar.ToString().ToUpper();
15    }
  1 reference
16    public void t1_textchanged(object sender, EventArgs e)
17    {
18        t2.Text = t1.Text.ToUpper() ;
19    }
  1 reference
20    public void t3_onfocus(object sender, EventArgs e)
21    {
22        t3.BackColor = Color.FromArgb(223, 223, 250);
23    }
  1 reference
24    public void t3_onleave(object sender, EventArgs e)
25    {
26        t3.BackColor = Color.White;
27    }
  1 reference
28    public void t3_keydown(object sender, System.Windows.Forms.KeyEventArgs e)
29    {
30        if (e.Control == true && e.KeyCode == Keys.Home)
31        {
32            t3.Text = "Clarent Institue";
33        }
34    }
  1 reference
35    public void t4_keyup(object sender, System.Windows.Forms.KeyEventArgs e)
36    {
37        if (e.Control == true && e.Shift == true && e.KeyCode == Keys.End)
38        {
39            t4.Text = "Case Point Intern";
40        }
41    }
  1 reference
42    public void t4_mouseenter(object sender, EventArgs e)
43    {
44        t4.BackColor = Color.FromArgb(225, 250, 225);
45    }

```

```
46 public void t4_mouseleave(object sender, EventArgs e)
```

```
47 {  
48     t4.BackColor = Color.White;  
49 }
```

```
1 reference
```

```
50 public void b1_mouseclick(object sender, MouseEventArgs e)
```

```
51 {  
52     MessageBox.Show($" Button {e.Button} clicked ");  
53 }
```

```
1 reference
```

```
54 public void p1_mousedown(object sender, MouseEventArgs e)
```

```
55 {  
56     startPoint = e.Location; // Capture start position  
57     isDrawing = true;  
58 }
```

```
1 reference
```

```
59 public void p1_mouseup(object sender, MouseEventArgs e)
```

```
60 {  
61     if (isDrawing)  
62     {  
63  
64         isDrawing = false;  
65         using (Graphics g = Graphics.FromImage(canvas))  
66         {  
67             Pen pen = new Pen(Color.Black, 2); // Black line with width 2  
68             g.DrawLine(pen, startPoint, e.Location);  
69         }  
70         p1.Image = canvas;  
71         p1.Refresh();  
72     }  
73 }  
74  
75 }  
76
```

```

18 private void InitializeComponent()
19 {
20     this.comboBoxExample = new System.Windows.Forms.ComboBox();
21     this.numericUpDownExample = new System.Windows.Forms.NumericUpDown();
22     this.groupBoxExample = new System.Windows.Forms.GroupBox();
23     this.radioButtonFemale = new System.Windows.Forms.RadioButton();
24     this.(field) CheckBox FrmControl.checkBoxExample Button();
25     this.checkBoxExample = new System.Windows.Forms.CheckBox();
26     this.btnShow = new System.Windows.Forms.Button();
27     this.lblResult = new System.Windows.Forms.Label();
28     //
29     // comboBoxExample
30     //
31     this.comboBoxExample.FormattingEnabled = true;
32     this.comboBoxExample.Location = new System.Drawing.Point(30, 30);
33     this.comboBoxExample.Name = "comboBoxExample";
34     this.comboBoxExample.Size = new System.Drawing.Size(150, 24);
35     this.comboBoxExample.TabIndex = 0;
36     //
37     // numericUpDownExample
38     //
39     this.numericUpDownExample.Location = new System.Drawing.Point(30, 70);
40     this.numericUpDownExample.Maximum = 100;
41     this.numericUpDownExample.Name = "numericUpDownExample";
42     this.numericUpDownExample.Size = new System.Drawing.Size(150, 22);
43     this.numericUpDownExample.TabIndex = 1;
44     //
45     // groupBoxExample
46     //
47     this.groupBoxExample.Controls.Add(this.radioButtonFemale);
48     this.groupBoxExample.Controls.Add(this.radioButtonMale);
49     this.groupBoxExample.Location = new System.Drawing.Point(30, 110);
50     this.groupBoxExample.Name = "groupBoxExample";
51     this.groupBoxExample.Size = new System.Drawing.Size(200, 70);
52     this.groupBoxExample.TabIndex = 2;
53     this.groupBoxExample.TabStop = false;
54     this.groupBoxExample.Text = "Choose Gender";
55     //
56     // radioButtonFemale
57     //
58     this.radioButtonFemale.AutoSize = true;
59     this.radioButtonFemale.Location = new System.Drawing.Point(100, 30);
60     this.radioButtonFemale.Name = "radioButtonFemale";
61     this.radioButtonFemale.Size = new System.Drawing.Size(75, 21);
62     this.radioButtonFemale.TabIndex = 1;
63     this.radioButtonFemale.TabStop = true;
64     this.radioButtonFemale.Text = "Female";
65     this.radioButtonFemale.UseVisualStyleBackColor = true;

```

```

67 // radioButtonMale
68 //
69 this.radioButtonMale.AutoSize = true;
70 this.radioButtonMale.Location = new System.Drawing.Point(20, 30);
71 this.radioButtonMale.Name = "radioButtonMale";
72 this.radioButtonMale.Size = new System.Drawing.Size(59, 21);
73 this.radioButtonMale.TabIndex = 0;
74 this.radioButtonMale.TabStop = true;
75 this.radioButtonMale.Text = "Male";
76 this.radioButtonMale.UseVisualStyleBackColor = true;
77 //
78 // checkBoxExample
79 //
80 this.checkBoxExample.AutoSize = true;
81 this.checkBoxExample.Location = new System.Drawing.Point(30, 190);
82 this.checkBoxExample.Name = "checkBoxExample";
83 this.checkBoxExample.Size = new System.Drawing.Size(164, 21);
84 this.checkBoxExample.TabIndex = 3;
85 this.checkBoxExample.Text = "I agree to the terms";
86 this.checkBoxExample.UseVisualStyleBackColor = true;
87 //
88 // btnShow
89 //
90 this.btnShow.Location = new System.Drawing.Point(30, 230);
91 this.btnShow.Name = "btnShow";
92 this.btnShow.Size = new System.Drawing.Size(150, 30);
93 this.btnShow.TabIndex = 4;
94 this.btnShow.Text = "Show Selected";
95 this.btnShow.UseVisualStyleBackColor = true;
96 this.btnShow.Click += new System.EventHandler(this.btnShow_Click);
97 //
98 // lblResult
99 //
100 this.lblResult.AutoSize = true;
101 this.lblResult.Location = new System.Drawing.Point(30, 280);
102 this.lblResult.Name = "lblResult";
103 this.lblResult.Size = new System.Drawing.Size(46, 17);
104 this.lblResult.TabIndex = 5;
105 this.lblResult.Text = "Result";
106 //
107 // FrmControl
108 //
109 this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 16F);
110 this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
111 this.ClientSize = new System.Drawing.Size(400, 350);
112 this.Controls.Add(this.lblResult);
113 this.Controls.Add(this.btnShow);
114 this.Controls.Add(this.checkBoxExample);
115 this.Controls.Add(this.groupBoxExample);
116 this.Controls.Add(this.numericUpDownExample);
117 this.Controls.Add(this.comboBoxExample);
118 this.Name = "FrmControl";

```

```
119     this.Text = "Demo Controls";  
120     this.Load += new System.EventHandler(this.FrmControl_Load);  
121  
122 }  
123  
124  
125
```

13 references

```
126 private System.Windows.Forms.ComboBox comboBoxExample;
```

8 references

```
127 private System.Windows.Forms.NumericUpDown numericUpDownExample;
```

10 references

```
128 private System.Windows.Forms.GroupBox groupBoxExample;
```

11 references

```
129 private System.Windows.Forms.RadioButton radioButtonFemale;
```

12 references

```
130 private System.Windows.Forms.RadioButton radioButtonMale;
```

10 references

```
131 private System.Windows.Forms.CheckBox checkBoxExample;
```

9 references

```
132 private System.Windows.Forms.Button btnShow;
```

9 references

```
133 private System.Windows.Forms.Label lblResult;
```

```
134 }
```

```
135 }
```

```
136
```

```

9 namespace FormApp
10 {
11     2 references
12     public partial class FrmControl : Form
13     {
14         0 references
15         public FrmControl()
16         {
17             InitializeComponent();
18
19         1 reference
20         private void FrmControl_Load(object sender, EventArgs e)
21         {
22             comboBoxExample.DropDownStyle = ComboBoxStyle.DropDownList;
23             radioButtonMale.Checked = true;
24             // Initialize ComboBox items
25             comboBoxExample.Items.Add("Option 1");
26             comboBoxExample.Items.Add("Option 2");
27             comboBoxExample.Items.Add("Option 3");
28             comboBoxExample.SelectedIndex = 0; // Default selection
29
30         1 reference
31         private void btnShow_Click(object sender, EventArgs e)
32         {
33             string comboText = comboBoxExample.SelectedItem.ToString();
34             decimal numericValue = numericUpDownExample.Value;
35             string gender = radioButtonMale.Checked ? "Male" : radioButtonFemale.Checked ? "Female" : "Not selected";
36             string terms = checkBoxExample.Checked ? "Agreed" : "Not Agreed";
37
38             lblResult.Text = $"ComboBox: {comboText}\n" +
39                             $"NumericUpDown: {numericValue}\n" +
40                             $"Gender: {gender}\n" +
41                             $"Terms: {terms}";
42         }
43     }
}

```

```

14 private void InitializeComponent()
15 {
16     this.btnColor = new System.Windows.Forms.Button();
17     this.btnFont = new System.Windows.Forms.Button();
18     this.btnOpenFile = new System.Windows.Forms.Button();
19     this.btnSaveFile = new System.Windows.Forms.Button();
20     this.btnFolder = new System.Windows.Forms.Button();
21     this.textBoxResult = new System.Windows.Forms.TextBox();
22     this.SuspendLayout();
23
24     this.btnColor.Location = new System.Drawing.Point(20, 20);
25     this.btnColor.Name = "btnColor";
26     this.btnColor.Size = new System.Drawing.Size(150, 30);
27     this.btnColor.TabIndex = 0;
28     this.btnColor.Text = "Open color Dialog";
29     this.btnColor.UseVisualStyleBackColor = true;
30     this.btnColor.Click += new System.EventHandler(this.btnColor_Click);
31
32     this.btnFont.Location = new System.Drawing.Point(20, 60);
33     this.btnFont.Name = "btnFont";
34     this.btnFont.Size = new System.Drawing.Size(150, 30);
35     this.btnFont.TabIndex = 0;
36     this.btnFont.Text = "Open Font Dialog";
37     this.btnFont.UseVisualStyleBackColor = true;
38     this.btnFont.Click += new System.EventHandler(this.btnFont_Click);
39
40     this.btnOpenFile.Location = new System.Drawing.Point(20, 100);
41     this.btnOpenFile.Name = "btnOpenFile";
42     this.btnOpenFile.Size = new System.Drawing.Size(150, 30);
43     this.btnOpenFile.TabIndex = 1;
44     this.btnOpenFile.Text = "Open File Dialog";
45     this.btnOpenFile.UseVisualStyleBackColor = true;
46     this.btnOpenFile.Click += new System.EventHandler(this.btnOpenFile_Click);
47
48     this.btnSaveFile.Location = new System.Drawing.Point(20, 140);
49     this.btnSaveFile.Name = "btnSaveFile";
50     this.btnSaveFile.Size = new System.Drawing.Size(150, 30);
51     this.btnSaveFile.TabIndex = 2;
52     this.btnSaveFile.Text = "Save File Dialog";
53     this.btnSaveFile.UseVisualStyleBackColor = true;
54     this.btnSaveFile.Click += new System.EventHandler(this.btnSaveFile_Click);
55

```

```

56     this.btnFolder.Location = new System.Drawing.Point(20, 180);
57     this.btnFolder.Name = "btnFolder";
58     this.btnFolder.Size = new System.Drawing.Size(150, 30);
59     this.btnFolder.TabIndex = 3;
60     this.btnFolder.Text = "Open Folder Dialog";
61     this.btnFolder.UseVisualStyleBackColor = true;
62     this.btnFolder.Click += new System.EventHandler(this.btnFolder_Click);
63
64     this.textBoxResult.Location = new System.Drawing.Point(20, 230);
65     this.textBoxResult.Multiline = true;
66     this.textBoxResult.Name = "textBoxResult";
67     this.textBoxResult.ScrollBars = System.Windows.Forms.ScrollBars.Vertical;
68     this.textBoxResult.Size = new System.Drawing.Size(350, 150);
69     this.textBoxResult.TabIndex = 4;
70
71     this.ClientSize = new System.Drawing.Size(800, 370);
72     this.Controls.Add(this.textBoxResult);
73     this.Controls.Add(this.btnFolder);
74     this.Controls.Add(this.btnSaveFile);
75     this.Controls.Add(this.btnOpenFile);
76     this.Controls.Add(this.btnFont);
77     this.Controls.Add(this.btnColor);
78     this.Name = "FrmDialog";
79     this.Text = "Dialog Demonstration";
80     this.ResumeLayout(false);
81     this.PerformLayout();
82 }
83
84 private System.Windows.Forms.Button btnColor;
85
86 private System.Windows.Forms.Button btnFont;
87
88 private System.Windows.Forms.Button btnOpenFile;
89
90 private System.Windows.Forms.Button btnSaveFile;
91
92 private System.Windows.Forms.Button btnFolder;
93
94 private System.Windows.Forms.TextBox textBoxResult;
95

```

```

1 using System;
2 using System.Drawing;
3 using System.Windows.Forms;
4 namespace FormApp
5 {
6     2 references
7     public partial class FrmDialog : Form
8     {
9         0 references
10        public FrmDialog()
11        {
12            InitializeComponent();
13        }
14        1 reference
15        private void btnColor_Click(object sender, EventArgs e)
16        {
17            ColorDialog colorDialog = new ColorDialog();
18            if (colorDialog.ShowDialog() == DialogResult.OK)
19            {
20                this.textBoxResult.BackColor = colorDialog.Color;
21            }
22        }
23        1 reference
24        private void btnFont_Click(object sender, EventArgs e)
25        {
26            FontDialog fontDialog = new FontDialog();
27            if (fontDialog.ShowDialog() == DialogResult.OK)
28            {
29                this.textBoxResult.Font = fontDialog.Font;
30                this.textBoxResult.Text = "Selected Font: " + fontDialog.Font.Name + ", " + fontDialog.Font.Size.ToString();
31            }
32        }
33        1 reference
34        private void btnOpenFile_Click(object sender, EventArgs e)
35        {
36            OpenFileDialog openFileDialog = new OpenFileDialog();
37            openFileDialog.Filter = "Text Files|*.txt|All Files|*.*";
38            if (openFileDialog.ShowDialog() == DialogResult.OK)
39            {
40                this.textBoxResult.Text = "Opened File: " + openFileDialog.FileName;
41            }
42        }
43    }
44 }

```

```

38 private void btnSaveFile_Click(object sender, EventArgs e)
39 {
40     SaveFileDialog saveFileDialog = new SaveFileDialog();
41     saveFileDialog.Filter = "Text Files|*.txt|All Files|*.*";
42     if (saveFileDialog.ShowDialog() == DialogResult.OK)
43     {
44         this.textBoxResult.Text = "File to Save: " + saveFileDialog.FileName;
45     }
46 }
47
48 1 reference
49 private void btnFolder_Click(object sender, EventArgs e)
50 {
51     FolderBrowserDialog folderDialog = new FolderBrowserDialog();
52     if (folderDialog.ShowDialog() == DialogResult.OK)
53     {
54         this.textBoxResult.Text = "Selected Folder: " + folderDialog.SelectedPath;
55     }
56 }
57 }
58

```

```

22 private void InitializeComponent()
23 {
24     this.menuStripMain = new System.Windows.Forms.MenuStrip();
25     this.fileToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
26     this.sDIToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
27     this.sDIDialogToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
28     this.mDIToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
29     this.windowToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
30     this.tileHorizontalToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
31     this.tileVerticalToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
32     this.cascadeToolStripMenuItem = new System.Windows.Forms.ToolStripItem();
33
34
35     this.menuStripMain.ImageScalingSize = new System.Drawing.Size(20, 20);
36     this.menuStripMain.Items.AddRange(new System.Windows.Forms.ToolStripItem[] {
37         this.fileToolStripMenuItem,
38         this.windowToolStripMenuItem});
39     this.menuStripMain.Location = new System.Drawing.Point(0, 0);
40     this.menuStripMain.Name = "menuStripMain";
41     this.menuStripMain.Size = new System.Drawing.Size(800, 28);
42     this.menuStripMain.TabIndex = 0;
43
44     this.fileToolStripMenuItem.DropDownItems.AddRange(new System.Windows.Forms.ToolStripItem[] {
45         this.sDIToolStripMenuItem, this.sDIDialogToolStripMenuItem,
46         this.mDIToolStripMenuItem});
47     this.fileToolStripMenuItem.Name = "fileToolStripMenuItem";
48     this.fileToolStripMenuItem.Text = "File";
49
50     this.sDIToolStripMenuItem.Name = "sDIToolStripMenuItem";
51     this.sDIToolStripMenuItem.Text = "SDI Child";
52     this.sDIToolStripMenuItem.Click += new System.EventHandler(this.sDIToolStripMenuItem_Click);
53
54     this.sDIDialogToolStripMenuItem.Name = "sDIDialogToolStripMenuItem";
55     this.sDIDialogToolStripMenuItem.Text = "SDI Child Dialog";
56     this.sDIDialogToolStripMenuItem.Click += new System.EventHandler(this.sDIDialogToolStripMenuItem_Click);
57
58
59     this.mDIToolStripMenuItem.Name = "mDIToolStripMenuItem";
60     this.mDIToolStripMenuItem.Text = "MDI Child";
61     this.mDIToolStripMenuItem.Click += new System.EventHandler(this.mDIToolStripMenuItem_Click);

```

```

63 this.windowToolStripMenuItem.DropDownItems.AddRange(new System.Windows.Forms.ToolStripItem[] {
64     this.tileHorizontalToolStripMenuItem,
65     this.tileVerticalToolStripMenuItem,
66     this.cascadeToolStripMenuItem});
67 this.windowToolStripMenuItem.Name = "windowToolStripMenuItem";
68 this.windowToolStripMenuItem.Text = "Window";
69
70     this.tileHorizontalToolStripMenuItem.Name = "tileHorizontalToolStripMenuItem";
71     this.tileHorizontalToolStripMenuItem.Text = "Tile Horizontal";
72     this.tileHorizontalToolStripMenuItem.Click += new System.EventHandler(this.tileHorizontalToolStripMenuItem_Click);
73
74     this.tileVerticalToolStripMenuItem.Name = "tileVerticalToolStripMenuItem";
75     this.tileVerticalToolStripMenuItem.Text = "Tile Vertical";
76     this.tileVerticalToolStripMenuItem.Click += new System.EventHandler(this.tileVerticalToolStripMenuItem_Click);
77
78     this.cascadeToolStripMenuItem.Name = "cascadeToolStripMenuItem";
79     this.cascadeToolStripMenuItem.Text = "Cascade";
80     this.cascadeToolStripMenuItem.Click += new System.EventHandler(this.cascadeToolStripMenuItem_Click);
81
82     this.ClientSize = new System.Drawing.Size(800, 450);
83     this.Controls.Add(this.menuStripMain);
84     this.MainMenuStrip = this.menuStripMain;
85     this.MainMenuStrip.MdiWindowListItem = windowToolStripMenuItem;
86     this.Name = "FrmMain";
87     this.Text = "MDI / SDI Demo";
88 }

```

9 references
private System.Windows.Forms.MenuStrip menuStripMain;

5 references
private System.Windows.Forms.ToolStripItem fileToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem sDIToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem sDIDialogToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem mDIToolStripMenuItem;

6 references
private System.Windows.Forms.ToolStripItem windowToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem tileHorizontalToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem tileVerticalToolStripMenuItem;

5 references
private System.Windows.Forms.ToolStripItem cascadeToolStripMenuItem;

98 }

99 }

1 reference

```
private void InitializeComponent()  
{  
    this.textBoxChild = new System.Windows.Forms.TextBox();  
    this.SuspendLayout();  
    //  
    // textBoxChild  
    //  
    this.textBoxChild.Location = new System.Drawing.Point(20, 20);  
    this.textBoxChild.Multiline = true;  
    this.textBoxChild.Name = "textBoxChild";  
    this.textBoxChild.Size = new System.Drawing.Size(250, 150);  
    this.textBoxChild.TabIndex = 0;  
    //  
    // FrmChild  
    //  
    this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 16F);  
    this.ClientSize = new System.Drawing.Size(300, 200);  
    this.Controls.Add(this.textBoxChild);  
    this.Name = "FrmChild";  
    this.Text = "Child Form";  
    this.ResumeLayout(false);  
    this.PerformLayout();  
}
```

7 references

```
private System.Windows.Forms.TextBox textBoxChild;
```

1 using System;

2 using System.Collections.Generic;

3 using System.Linq;

4 using System.Threading.Tasks;

6 namespace FormApp

7 {

8 references

public partial class FrmChild : Form

{

3 references

public FrmChild()

{

InitializeComponent();

}

}

15 }

```

1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5
6 namespace FormApp
7 {
8     2 references
9     public partial class FrmMain : Form
10     {
11         0 references
12         public FrmMain()
13         {
14             InitializeComponent();
15             this.IsMdiContainer = true;
16         }
17
18         1 reference
19         private void sDIToolStripMenuItem_Click(object sender, EventArgs e)
20         {
21             FrmChild child = new FrmChild();
22             child.Text = "SDI Child";
23             child.Show();
24         }
25
26         1 reference
27         private void sDIDialogToolStripMenuItem_Click(object sender, EventArgs e)
28         {
29             FrmChild child = new FrmChild();
30             child.Text = "SDI Child";
31             child.ShowDialog();
32         }
33
34         1 reference
35         private void mDIToolStripMenuItem_Click(object sender, EventArgs e)
36         {
37             FrmChild child = new FrmChild();
38             child.MdiParent = this;
39             child.Text = "MDI Child";
40             child.Show();
41         }
42
43         1 reference
44         private void tileHorizontalToolStripMenuItem_Click(object sender, EventArgs e)
45         {
46             this.LayoutMdi(MdiLayout.TileHorizontal);
47         }
48
49         1 reference
50         private void tileVerticalToolStripMenuItem_Click(object sender, EventArgs e)
51         {
52             this.LayoutMdi(MdiLayout.TileVertical);
53         }
54
55         1 reference
56         private void cascadeToolStripMenuItem_Click(object sender, EventArgs e)
57         {
58             this.LayoutMdi(MdiLayout.Cascade);
59         }
60     }
61 }

```

```
6  namespace FormApp
7  {
    11 references
8  public class Employee
9  {
    6 references
10     public int EmpId { get; set; }
    6 references
11     public string EmpName { get; set; }
    6 references
12     public decimal Salary { get; set; }
    6 references
13     public int DeptNo { get; set; }
14 }
15
16 }
```

Emp ID

Add

Emp Name

Update

Salary

Delete

Dept No

View All

Col1

Col2

Col3

```

1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using Npgsql;
6 namespace FormApp
7 {
8     2 references
9     public class EmployeeService
10     {
11         5 references
12         private string connectionString = "Server=localhost;port=5432;Database=stud2025;User Id=postgres;Password=Admin@123;";
13
14         // Create
15         1 reference
16         public void AddEmployee(Employee emp)
17         {
18             var conn = new NpgsqlConnection(connectionString);
19             try
20             {
21                 conn.Open();
22                 using var cmd = new NpgsqlCommand("INSERT INTO t_employee(empid, empname, salary, deptno) VALUES (@id, @name, @salary, @deptno)", conn);
23                 cmd.Parameters.AddWithValue("id", emp.EmpId);
24                 cmd.Parameters.AddWithValue("name", emp.EmpName);
25                 cmd.Parameters.AddWithValue("salary", emp.Salary);
26                 cmd.Parameters.AddWithValue("deptno", emp.DeptNo);
27                 cmd.ExecuteNonQuery();
28             }
29             catch (Exception ex)
30             {
31                 MessageBox.Show("Error adding employee: " + ex.Message);
32             }
33             finally
34             {
35                 conn.Close();// Any cleanup code if necessary
36             }
37         }
38
39         // Read all
40         1 reference
41         public List<Employee> GetAllEmployees()
42         {
43             var list = new List<Employee>();
44             var conn = new NpgsqlConnection(connectionString);
45             conn.Open();
46             using var cmd = new NpgsqlCommand("SELECT empid, empname, salary, deptno FROM t_employee", conn);
47             using var reader = cmd.ExecuteReader();
48             while (reader.Read())
49             {
50             }
51         }
52     }
53 }

```

```

54 while (reader.Read())
55 {
56     list.Add(new Employee
57     {
58         EmpId = reader.GetInt32(0),
59         EmpName = reader.GetString(1),
60         Salary = reader.GetDecimal(2),
61         DeptNo = reader.IsDBNull(3) ? 0 : reader.GetInt32(3)
62     });
63 }
64 return list;
65 }
66
67 // Read single
68 0 references
69 public Employee GetEmployee(int empId)
70 {
71     try
72     {
73         using var conn = new NpgsqlConnection(connectionString);
74         conn.Open();
75         using var cmd = new NpgsqlCommand("SELECT empid, empname, salary, deptno FROM t_employee WHERE empid = @id", conn);
76         cmd.Parameters.AddWithValue("id", empId);
77         using var reader = cmd.ExecuteReader();
78         if (reader.Read())
79         {
80             return new Employee
81             {
82                 EmpId = reader.GetInt32(0),
83                 EmpName = reader.GetString(1),
84                 Salary = reader.GetDecimal(2),
85                 DeptNo = reader.IsDBNull(3) ? 0 : reader.GetInt32(3)
86             };
87         }
88     }
89     catch (Exception ex)
90     {
91         MessageBox.Show("Error retrieving employee: " + ex.Message);
92     }
93     return null;
94 }
95
96 // Update
97 1 reference
98 public bool UpdateEmployee(Employee emp)
99 {
100     try
101     {
102         using var conn = new NpgsqlConnection(connectionString);
103         conn.Open();
104         using var cmd = new NpgsqlCommand("UPDATE t_employee SET empname=@name, salary=@salary, deptno=@deptno WHERE empid=@id", conn);
105         cmd.Parameters.AddWithValue("id", emp.EmpId);
106         cmd.Parameters.AddWithValue("name", emp.EmpName);
107         cmd.Parameters.AddWithValue("salary", emp.Salary);
108         cmd.Parameters.AddWithValue("deptno", emp.DeptNo);
109         int affected = cmd.ExecuteNonQuery();
110         return affected > 0;
111     }
112     catch (Exception ex)
113     {
114         MessageBox.Show("Error updating employee: " + ex.Message);
115         return false;
116     }
117 }
118
119 // Delete
120 1 reference
121 public bool DeleteEmployee(int empId)
122 {
123     try
124     {
125         using var conn = new NpgsqlConnection(connectionString);
126         conn.Open();
127         using var cmd = new NpgsqlCommand("DELETE FROM t_employee WHERE empid = @id", conn);
128         cmd.Parameters.AddWithValue("id", empId);
129         int affected = cmd.ExecuteNonQuery();
130         return affected > 0;
131     }
132     catch (Exception ex)
133     {
134         MessageBox.Show("Error deleting employee: " + ex.Message);
135         return false;
136     }
137 }
138 }
139 }
140 }
141 }
142 }
143 }
144 }
145 }

```

```

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5  using System;
6  using System.Windows.Forms;
7
8  namespace FormApp
9  {
10     2 references
11     public partial class FrmEmployee : Form
12     {
13         4 references
14         private EmployeeService _service = new EmployeeService();
15         0 references
16         public FrmEmployee()
17         {
18             InitializeComponent();
19             LoadEmployees();
20         }
21         1 reference
22         private void btnAdd_Click(object sender, EventArgs e)
23         {
24             try
25             {
26                 Employee emp = new Employee
27                 {
28                     EmpId = int.Parse(txtEmpId.Text),
29                     EmpName = txtEmpName.Text,
30                     Salary = decimal.Parse(txtSalary.Text),
31                     DeptNo = int.Parse(txtDeptNo.Text)
32                 };
33                 _service.AddEmployee(emp);
34                 MessageBox.Show("Employee added successfully!");
35                 LoadEmployees();
36                 ClearFields();
37             }
38             catch (Exception ex)
39             {
40                 MessageBox.Show("Error: " + ex.Message);
41             }
42         }

```

```

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79
1 reference
private void btnUpdate_Click(object sender, EventArgs e)
{
    try
    {
        Employee emp = new Employee
        {
            EmpId = int.Parse(txtEmpId.Text),
            EmpName = txtEmpName.Text,
            Salary = decimal.Parse(txtSalary.Text),
            DeptNo = int.Parse(txtDeptNo.Text)
        };
        bool updated = _service.UpdateEmployee(emp);
        if (updated)
        {
            MessageBox.Show("Employee updated successfully!");
            LoadEmployees();
            ClearFields();
        }
        else
        {
            MessageBox.Show("Employee not found.");
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}
1 reference
private void btnDelete_Click(object sender, EventArgs e)
{
    try
    {
        int empId = int.Parse(txtEmpId.Text);
        bool deleted = _service.DeleteEmployee(empId);
        if (deleted)
        {
            MessageBox.Show("Employee deleted successfully!");
            LoadEmployees();
            ClearFields();
        }
    }
}

```

```

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110
111
else
{
    MessageBox.Show("Employee not found.");
}
}
catch (Exception ex)
{
    MessageBox.Show("Error: " + ex.Message);
}
}
1 reference
private void btnView_Click(object sender, EventArgs e)
{
    LoadEmployees();
}
5 references
private void LoadEmployees()
{
    dgvEmployees.DataSource = null;
    dgvEmployees.DataSource = _service.GetAllEmployees();
}
3 references
private void ClearFields()
{
    txtEmpId.Clear();
    txtEmpName.Clear();
    txtSalary.Clear();
    txtDeptNo.Clear();
}
}
}

```

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5
6  namespace FormApp
7  {
8      10 references
9      public class Post
10     {
11         3 references
12         public int UserId { get; set; }
13         2 references
14         public int Id { get; set; }
15         3 references
16         public string? Title { get; set; }
17         3 references
18         public string? Body { get; set; }
19     }
20 }
```

```

1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Threading.Tasks;
5 using System.Net.Http.Json;
6 using System.Net;
7 namespace FormApp
8 {
9     3 references
10    public class ApiService
11    {
12        6 references
13        private readonly HttpClient _httpClient;
14        1 reference
15        public ApiService(string baseUrl)
16        {
17            _httpClient = new HttpClient { BaseAddress = new Uri(baseUrl) };
18        }
19        1 reference
20        public async Task<List<T>> GetAllAsync<T>(string endpoint)
21        {
22            return await _httpClient.GetFromJsonAsync<List<T>>(endpoint);
23        }
24        1 reference
25        public async Task<T> GetAsync<T>(string endpoint, int id)
26        {
27            return await _httpClient.GetFromJsonAsync<T>($" {endpoint}/{id}");
28        }
29        1 reference
30        public async Task<string> CreateAsync<T>(string endpoint, T item)
31        {
32            var response = await _httpClient.PostAsJsonAsync(endpoint, item);
33            return ((int)response.StatusCode).ToString();
34        }
35        1 reference
36        public async Task<string> UpdateAsync<T>(string endpoint, int id, T item)
37        {
38            var response = await _httpClient.PutAsJsonAsync($" {endpoint}/{id}", item);
39            return ((int)response.StatusCode).ToString();
40        }
41        1 reference
42        public async Task<string> DeleteAsync(string endpoint, int id)
43        {
44            var response = await _httpClient.DeleteAsync($" {endpoint}/{id}");
45            return ((int)response.StatusCode).ToString();
46        }
47    }
48 }

```

0	send	id	title	body
1	1	1	sunt aut facere ...	quia et suscipit...
1	2	2	qui est esse	est rerum temp...
1	3	3	ea molestias qu...	et iusto sed qu...
1	4	4	eum et est occa...	ullam et saepe r...
1	5	5	nesciunt quas o...	repudiandae ve...
1	6	6	dolorem eum ...	ut aspernatur c...
1	7	7	magnam facilis...	dolore placeat ...
1	8	8	dolorem dolore...	dignissimos ap...
1	9	9	nesciunt iure o...	consectetur ani...
1	10	10	optio molestias...	quo et expedita...
2	11	11	et ea vero quia l...	delectus reicien...
2	12	12	in quibusdam t...	itaque id aut m...
2	13	13	dolorum ut in v...	aut dicta possi...
2	14	14	voluptatem elig...	fuga et accusa...
2	15	15	eveniet quod te...	reprehenderit q...

Add

Update

Delete

Read

User Id

Title

Body

```
1 using System.Data;
2 namespace FormApp
3 {
4     3 references
5     public partial class Api : Form
6     {
7         6 references
8         private readonly ApiService _apiService;
9         2 references
10        private List<Post> _posts;
11        1 reference
12        public Api()
13        {
14            _apiService = new ApiService("https://jsonplaceholder.typicode.com/");
15            InitializeComponent();
16            LoadDataAsync();
17        }
18        1 reference
19        private async Task LoadDataAsync()
20        {
21            _posts = await _apiService.GetAllAsync<Post>("posts");
22            dgview.DataSource = _posts;
23        }
24        1 reference
25        public async void btnRead_Click(object sender, EventArgs e)
26        {
27            try
28            {
29                Post _post = await _apiService.GetAsync<Post>("posts", Int32.Parse(txtid.Text));
30                txtuserid.Text = Convert.ToString(_post.UserId);
31                txttitle.Text = _post.Title;
32                txtbody.Text = _post.Body;
33            }
34            catch (Exception ex)
35            {
36                MessageBox.Show("Please enter valid data: " + ex.Message, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
37            }
38        }
39        1 reference
40        public async void btnAdd_Click(object sender, EventArgs e)
41        {
42            try
43            {
44                Post post = new Post();
45                post.Id = int.Parse(txtid.Text);
46                post.UserId = int.Parse(txtuserid.Text);
47                post.Title = txttitle.Text;
48                post.Body = txtbody.Text;
49                string StatusCode = await _apiService.CreateAsync("posts", post);
50                if (StatusCode == "201")
51                {
52                    MessageBox.Show("Created sucessfully " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Information);
53                }
54                else
55                {
56                    MessageBox.Show("Not created some error " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
57                }
58            }
59            catch (Exception ex)
60            {
61                MessageBox.Show("Please enter valid data: " + ex.Message, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
62            }
63        }
64        1 reference
65        public async void btnUpdate_Click(object sender, EventArgs e)
66        {
67            try
68            {
69                Post post = new Post();
70                post.Id = int.Parse(txtid.Text);
71                post.UserId = int.Parse(txtuserid.Text);
72                post.Title = txttitle.Text;
73                post.Body = txtbody.Text;
74                string StatusCode = await _apiService.UpdateAsync("posts", Int32.Parse(txtid.Text), post);
75                if (StatusCode == "200")
76                {
77                    MessageBox.Show("Updated sucessfully " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Information);
78                }
79                else
80                {
81                    MessageBox.Show("Not updated some error " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
82                }
83            }
84            catch (Exception ex)
85            {
86                MessageBox.Show("Please enter valid data: " + ex.Message, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
87            }
88        }
89        1 reference
90        public async void btnDelete_Click(object sender, EventArgs e)
91        {
92            try
93            {
94                Post post = new Post();
95                string StatusCode = await _apiService.DeleteAsync("posts", Int32.Parse(txtid.Text));
96                if (StatusCode == "200")
97                {
98                    MessageBox.Show("Deleted sucessfully " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Information);
99                }
100               else
101               {
102                   MessageBox.Show("Not deleted some error " + StatusCode, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
103               }
104           }
105           catch (Exception ex)
106           {
107               MessageBox.Show("Please enter valid data: " + ex.Message, "Api", MessageBoxButtons.OK, MessageBoxIcon.Error);
108           }
109       }
110   }
111 }
```